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Perspective

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Fluorinated compounds in paper and paperboard based food packaging materials

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Paper- and paperboard-based materials are alternatives to petroleum-based plastics in food packaging but unsuitable for their poor moisture and oil resistance. In this sense, fluorinated compounds improve water and grease repellency, though their use is controversial. This Perspective discusses main techniques to combine fluorinated compounds with paper and paperboard, including water and oil contact angles and grease resistance values, and summarizes main legal aspects in Europe and the United States.

Global population has been steadily increasing over the years, projected to reach 9.7 billion by 2050[1]. This demographic trend presents substantial challenges for the food industry, demanding solutions for food security, waste reduction, and food quality assurance. In this context, food packaging plays a pivotal role by preserving the quality and safety of food products, protecting them against physical damage, microbial contamination, and oxidative deterioration, extending the shelf life of foods, reducing food wastes, and maintaining nutritional value, flavor, and texture[2].

Currently, food packaging encompasses a wide range of different materials such as petroleum-based plastics, paper- and paperboard-based materials, metals, and glasses. The selection of a specific packaging material depends on several factors such as the type of food, shelf life, environmental considerations, and regulatory requirements[3]. The most commonly used food packaging materials are petroleum-based plastics mainly because their exceptional physical properties, light weight, high chemical inertness, and low production costs. The massive and continuous consumption of these materials has led to a global pollution issue associated with the extraction of the raw materials for their production, fabrication, and management as wastes[4].

Cellulose-based materials, mainly paper (thin sheets produced by processing natural cellulose fibers) and paperboard (heavyweight, thicker papers typically used as containers) can be used as primary (in direct contact with food) or secondary (not in direct contact with food) food packaging. They comprise the 31 wt.-% of the global packaging market segment[3]. Their main advantages are low cost, renewability, convenience during storage and consumption, possibility to include consumers information and marketing, safety for the environment, recyclability, and biodegradability after use[5]. Despite these advantages, the hydrophilic nature of cellulose and the fiber network porosity confer to paper and paperboard some important drawbacks such as low barrier properties and resistance to oil and greases, heat-sealability, and strength. For these reasons, paper and paperboard are usually combined with other materials in order to improve their functionality. In fact, of all primary food packaging, paper and paperboard account for a 20 wt.-%, and only a 3 wt.-% is used without further treatments[6].

In this Perspective, the key strategies related to the use of fluorinated compounds in food packaging to improve properties of paper and paperboard have been described. Main fabrication techniques, human health and environmental issues derived from their extensive use, classification, and legal aspects have been also presented and discussed.

Incorporation of substances to paper and paperboard to improve their performances

A wide range of materials can be combined with paper and paperboard to improve their functionality. Petroleum-based plastics and aluminum are the most used to enhance barrier properties even if their use results in a lack of biodegradability after use[5]. But with the target on biodegradability, bio-based plastics have emerged as a promising option. In this sense, different polysaccharides[7], polyesters[8], and proteins[9] have been described in the literature to effectively enhance paper and paperboard's properties. However, most of these solutions are expensive in comparison with petroleum-based plastics and none of them are extensively used in the market. Another strategy to make paper and paperboard suitable for food packaging applications is the use of fluorinated compounds. Such category comprises a very broad group of molecules and polymers containing at least one fluorine atom. The food packaging is a sector with a high demand of these substances, for example consuming ~33 wt.-% of the European production[6]. In general, fluorinated molecules present exceptional physical properties and, when used in combination with paper and paperboard, they confer interesting properties for food packaging applications such as water, oil, and stain resistance, non-flammability, high chemical and thermal stability, resistance to hydrolysis, photolysis and microbial degradation, as well as capacity to act as levelling agents (chemical substances added to increase the uniformity of

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a coating)[6,10–12], Table 1. Notwithstanding these properties, the use of fluorinated compounds, especially those having fluorinated chains ≥C8 and those with all hydrogens substituted by fluorine atoms (perfluoro compounds), implies their dispersion and accumulation in the environment as a consequence of their high stability and persistence, and no possibility of assimilation by living organisms and high risk of accumulation[10]. For this reason, some of them are the so-called "forever chemicals". To overcome this problem, academia and industry are focused on the development of less polluting alternatives such as those with shorter chains (≤C6) and on the attachment of fluorinated chains to the backbones of biodegradable molecules[13]. In this sense, polysaccharides represent a viable candidate due to their complete biodegradability and the possibility of functionalization through chemical reactions with their free hydroxyl groups, being the esterification and the etherification the most used[14]. With such methodology, fluorinated polysaccharide derivatives can be obtained by reaction with fluorinated compounds[15], Fig. 1.

Table 1 | Main advantages and disadvantages associated with the use of fluorinated compounds in food packaging

Advantages	Disadvantages
Water and oil repellency	Manufacturing pollution
Non-flammability	Biopersistance
Chemical stability	Not biodegradable
Thermal stability	Bioaccumulation
Resistance to hydrolysis, photolysis, and microbial degradation	
Capacity to act as levelling agents	

Classification of fluorinated compounds used in food packaging

Since the 1950s, a wide variety of fluorinated substances have been synthesized, reaching over 4700 compounds on the global market, of which less than 6 wt.-% are of commercial interest. Some of them are illustrated in Fig. 2A. This number is much higher (~8000) if following the definition of the European Chemicals Agency by which a fluorinated substance is any compound containing at least one -CF_3 or -CF_2- group[11,16,17]. In this scenario, a single, clear, and agreed classification of these synthetic compounds is needed for a better understanding for industrial and research sectors and to prevent the mismatch between different documents and reports[18], Fig. 2B.

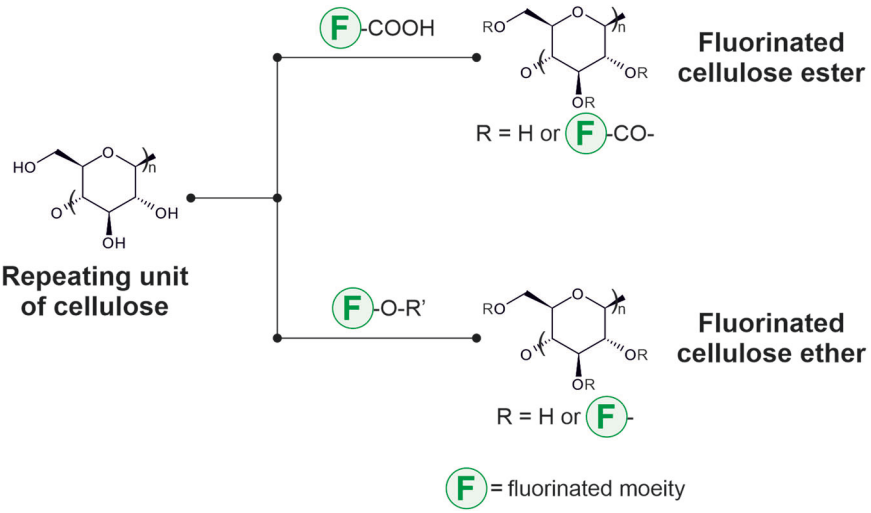


Fig. 1 | Cellulose functionalization. Scheme of two of the main reactions (i.e., esterification and etherification) for the chemical functionalization of cellulose with fluorinated molecules.

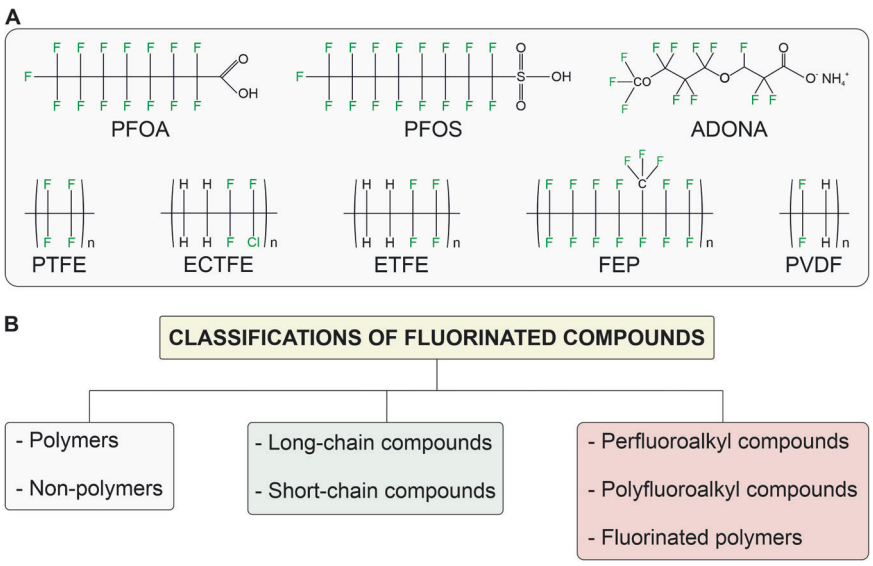


Fig. 2 | Examples and classification of fluorinated molecules and polymers. A chemical structures of some fluorinated molecules (top) such as perfluorooctanoic acid (PFOA), perfluorooctane sulfonic acid (PFOS), and 3H-perfluoro-3-[(3-methoxy-propoxy)propanoic acid] (ADONA), and fluoropolymers of commercial interest (bottom) such as polytetrafluoroethylene (PTFE), polyethylene chlorotrifluoroethylene (ECTFE), polyethylene tetrafluoroethylene (ETFE), fluorinated-ethylene-propylene (FEP), and polyvinylidene fluoride (PVDF). **B** schematics of some of the different classifications of fluorinated compounds.

Some authors propose a general classification distinguishing among polymers and non-polymers (mainly residues of fabrication, degradation products, or chemicals used as sizing agents in the production of water and oil repellent papers and paperboards) based on the special mobility of non-polymeric fluorinated substances and the enhanced possibility of migration if in contact with food[19]. Others suggest to focus in the length of the fluorinated chain, independently of the number of fluorine atoms or the chemical functionalization[11]. This is a controversial classification mostly due to the different definitions of "long-chain" and "short-chain" fluorinated molecules available in literature[18,20]. Other attempts to classify the different fluorinated molecules make the distinction in the number of fluorinated atoms and the chemical nature, resulting in three main categories: perfluoroalkyl compounds (all hydrogen atoms substituted with fluorine atoms), polyfluoroalkyl substances (at least one position is not substituted by fluorine atoms), and fluorinated polymers (chemical compounds consisting of many repeating units containing carbon-fluoride bonds)[11]. Finally, there are some authors stating that every single fluorinated substance is unique due to its physical properties, resulting in classifications with many categories and sub-categories and, hence, difficult to manage[17].

Parameters for the characterization of hydrophobicity, oleophobicity and oil/grease resistance of food packaging and the role of fluorinated molecules

The most employed parameter to determine the hydrophobicity of food packaging materials is the static water contact angle. For this, a water drop is applied on their surface and the angle between the liquid and the solid surface at the contact point is measured. For hydrophilic materials, water contact angle values are <90° whereas hydrophobic materials are characterized by contact angles >90°. Moreover, this versatile technique allows the determination of the oleophobicity by using liquids with different polarities and viscosities such as animal and vegetable oils. In general, oils tend to spread more than water on a surface, thus, oil contact angles are usually lower than those obtained by using water. However, there is no a clear definition of the contact angle value above which a material is considered oleophobic[21], with the general rule being that the higher the contact angle, the greater the oleophobicity. Independently of the liquid, the value of the contact angle is influenced by different factors such as the roughness, chemical nature, and heterogeneity of the solid surface, making difficult the direct comparison between literature data[21]. Typical water and oil contact angle values of different food packaging materials such as petroleum-based plastics, thermoplastic biopolymers, polysaccharide-based, protein-based, and fluorinated compounds are compiled in Tables 2 and 3. In general, for water contact angle, the lowest values are ascribed to polysaccharides mainly due to the high number of hydroxyl groups able to interact with water. In contrast, the fluorinated molecules show water contact angles >90° as a consequence of a lower interaction with the liquid drop, rendering the surface more hydrophobic and, thus, more resistant to water and moisture. For oil contact angles, there is an important lack of information in literature and no oil has been stated as a standard. In fact, by using diverse oils, variations of ~20° have been reported[22]. Even considering these differences, petroleum-based plastics show lower values than fluorinated compounds, being the higher values those ascribed to polytetrafluoroethylene (PTFE) and ethylene-tetrafluoroethylene copolymer (ETFE), Table 2.

On the other hand, grease resistance, commonly known as the kit test, is usually determined by following the Tappi test method 559 cm-22[4]. This is an inexpensive, standard method for the determination of repellency of paper and paperboard to grease, oil, and waxes consisting on the application of different solutions (mixtures of castor oil, toluene, and heptane numbered from 1 to 12) on the paper' surface followed by visual inspection after a certain time. The highest numbered solution applied on the surface leaving no dark spot is reported as the kit test value. For food packaging applications, this value should be >5. In general, virgin papers do not offer grease resistance. Cardboards, typically used as secondary food packaging materials, show good resistance with kit test values ~8, guaranteeing their functionality during transport and storage. For coated papers, the best values are ascribed to the use of fluorinated compounds (reaching values of 10–11) even if good values can be obtained by applying several layers of biopolymers such as chitosan (kit test value of 8), though resulting in less competitive prices and higher time-consuming processes. Table 3 reports oil contact angle and kit test values for non-treated papers and cardboards as well as for coated papers.

Table 2 | Typical water contact angle values of selected polymers used in food packaging applications

WATER WETTABILITY		
	Water contact angle (°)	References
Petroleum-based plastics		
High density polyethylene (HDPE)	75	36
Polyethylene terephthalate (PET)	81	21
Low density polyethylene (LDPE)	95	21
Polystyrene (PS)	95	37
Polypropylene (PP)	111	38
Thermoplastic food packaging biopolymers		
Thermoplastic starch (TPS)	32	39
Poly(butylene succinate) (PBS)	74	40
Poly(hydroxybutyrate) (PHB)	85	41
Polylactic acid (PLA)	92	42
Polysaccharide-based food packaging		
Cellulose paper	-*	43
Carrageenan	49	44
Alginate	60	45
Pectin	70	46
Chitosan	83	47
Starch	84	48
Chitin	91	4
Protein-based food packaging		
Corn zein	53	49
Soy protein	62	50
Collagen	103	51
Gelatin	105	49
Fluorinated food packaging		
Polyvinylidene fluoride (PVDF)	80	28
Ethylene-chlorotrifluoroethylene copolymer (ECTFE)	99	28
Ethylene-tetrafluoroethylene copolymer (ETFE)	108	28
Polyperfluoroalkoxyethylene (PFA)	121	28
Polytetrafluoroethylene (PTFE)	122	28
Coated papers		
C6-C16 fatty acids coated paper	66-87	52
Chitosan coated paper	95	53
Pentafluoroethane coated paper	115	54
Perfluoropolyether coated paper	121-140	55
PolyFox fluoromonomer coated paper	129-143	56
Perfluorononanoyl ester coated paper	161	57

(*) Water absorbing during the measurement time.

Fabrication techniques to improve paper and paperboard performance by using fluorinated compounds

Independently of the choice about the most convenient compound to combine with paper and paperboard, another important aspect to consider

is the fabrication method. Food packaging paper industry has developed different techniques to improve paper and paperboard functionality, being coating, lamination, and internal and external sizing the most used, Fig. 3.

Coating consists of the application of an external layer acting as a barrier for the paper matrix, Fig. 3A. These coatings, typically composed of pigments, binders, and additives, are applied on the paper and paperboard surface to achieve specific characteristics such as smoothness, printability, opacity, and ink adhesion. Furthermore, coatings can impart functionalities such as water and grease resistance and barrier properties, expanding the potential applications of paper and paperboard products by the addition of fluorinated molecules[3]. Lamination involves the application of several thin protective layers (often composed by different polymers) on the surface of the paper and paperboard, typically using adhesives, heat, and pressure, Fig. 3B. This strategy offers remarkable benefits such as moisture and tear resistance, light protection, and improved handling. Fluorinated compounds such as PTFE and ECTFE have been described to improve the water and vapor barrier of laminated packaging[23]. Moreover, lamination reinforces the structural integrity of paper and paperboard, rendering them suitable for the conservation of liquid food, otherwise impossible to package with cellulose derivatives. Finally, this process allows for the incorporation of specialized features like gloss or matte finishes contributing to product differentiation and brand recognition on retail shelves. The third main strategy to improve paper and paperboard properties is sizing. This is a fundamental process that significantly influences the properties and performance of the final product. It involves the application of specialty chemicals, known as sizing agents, to the paper fibers both to the paper slurry during manufacturing (internal sizing), Fig. 3C, or as an external layer, Fig. 3D. Fluorinated molecules such as fluorinated polyurethanes have been largely used by the food industry as sizing agents. These agents serve mainly to control the paper and paperboard's absorption and to improve its resistance to water and oil penetration, ensuring that ink and other liquids applied to the paper and paperboard remain on the surface, and reducing the risk of warping caused by moisture absorption[24]. Apart of these techniques, the paper industry is continuously searching for more environmentally friendly, faster, cost-effective, and scalable techniques. In this sense, chromatogeny has been recently proposed as an innovative solvent free grafting process that confers hydrophobicity to papers and boards in short times and using minimal volumes of reagents[25].

Table 3 | Typical oil contact angle and kit test values of selected polymers used in food packaging applications

OIL WETTABILITY		
	Oil contact angle (°)	**References**
Petroleum-based plastics		
High density polyethylene (HDPE)	38 (corn oil)	58
Polyethylene terephthalate (PET)	55 (ethylene glycol)	21
Low density polyethylene (LDPE)	68 (ethylene glycol)	21
Polystyrene (PS)	31 (hexadecane)	59
Polypropylene (PP)	57 (vegetable oils)	60
Fluorinated food packaging		
Polyvinylidene fluoride (PVDF)	52 (diiodomethane)	61
Ethylene-chlorotrifluoroethylene copolymer (ECTFE)	11 (hexadecane)	62
Ethylene-tetrafluoroethylene copolymer (ETFE)	152 (ethylene glycol)	63
Polyperfluoroalkoxyethylene (PFA)	54 (hexadecane)	62
Polytetrafluoroethylene (PTFE)	88 (diiodomethane)	64
GREASE RESISTANCE		
	Kit test value	**Reference**
Papers and cardboards		
Cellulose paper	0	43
Glassine paper	0	65
Noncandelered paper	1	66
Cardboard	8	67
Coated papers		
Pentafluoroethane coated paper	3–5	54
C6-C16 fatty acids coated paper	6–12	68
PolyFox fluoromonomer coated paper	7–8	56
Chitosan coated paper	8	68
Fluorinated resin coated paper	8–9	69
Perfluoropolyether coated paper	10–11	55

Health and environmental issues of the use of fluorinated compounds

The extensive use of fluorinated substances for different applications has led to a global widespread of these substances in the environment, wildlife, and humans mainly because their extremely low reactivity[20]. One of the main issues of these substances is their bioaccumulation, thus persisting for extended periods without degrading[26]. Their accumulation in living organisms can be turned into a biomagnification through the food chain, risking the wildlife and humans, as high concentrations of these compounds have been pointed out as disruptors and potentially carcinogen[27]. Additionally, these compounds can breakdown resulting in smaller molecules such as perfluorooctanoic acid, which is highly persistent. It has been largely described in literature for its ubiquity in landfills and water bodies[28] not only because its fragmentation but also due to its use as processing aids or during the synthesis of fluorinated compounds. Finally, the stability and low reactivity of fluorinated compounds make possible their long-range transport through air and water, leading to widespread distribution and potential contamination in regions far from their original sources[29].

When used in contact with foods, not only food packaging but also food contact materials (all materials and articles intended to come into contact with food at any stage of the production, processing, storage, preparation or consuming) can contribute to indirect dietary exposure thorough migration[10]. Migration refers to the exchange of chemical compounds between the packaging and the food by contact. The migration of

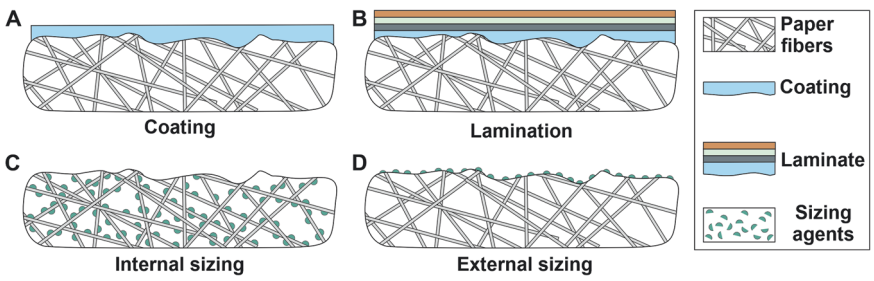


Fig. 3 | Methodologies to functionalize paper and paperboard. A–D, schematics of the different methodologies to improve paper and paperboard functionality: coating, lamination, and internal and external sizing, respectively.

fluorinated compounds into the foods is influenced by many factors[10,11,16]. Among them, the chemical composition of the packaging material, the specific type of fluorinated substance, its chain length, and the initial quantity used during the packaging fabrication. Typically, short chain fluorinated compounds such as those used as sizing agents, present a higher risk of migration mainly due to its higher mobility and the lack of a strong linkage (covalent, hydrogen bonds, etc.) with the cellulose fibers' network of papers and paperboards. Temperature, pH, and moisture of the environment and the protected food influence the migration process. Temperature is a parameter of particular interest in the case of fast food, usually wrapped as soon as prepared, and highly consumed in some western countries. Furthermore, the duration of the contact between the packaging and food as well as the type of food (e.g., fatty or aqueous) affect migration rates. For instance, fatty foods with a higher oil content, tend to interact more extensively with fluorinated compounds, leading to higher migration values[10]. Finally, mechanical stress and physical abrasion of the packaging associated with storage and transporting conditions may facilitate the release, resulting in a higher risk of food contamination.

Legal regulations of fluorinated compounds in food packaging

Similarly to classification, legislation about the use of fluorinated compounds in food packaging is not clear. It is mainly focused on long-chain perfluorinated compounds and it is scarce on other fluorine-containing substances. There is not a global regulation yet but different regulations for every region. The only ongoing initiative for a global legislation of plastics (including the use of fluorinated substances) is the United Nations Global Plastics Treaty, currently under negotiation, whose binding agreement is expected to be ready by 2024[30].

In Europe, legislation regarding the use of fluorinated substances in food packaging is primarily governed by the Framework Regulation (EC) No 1935/2004, which establishes the general principles for ensuring the safety of materials and articles intended for food contact[31]. Additionally, the Commission Regulation (EU) No 10/2011 specifically addresses plastic materials and articles intended to come into contact with food, including those containing fluorinated compounds and sets specific migration limits. The Commission Regulation (EC) No 2023/2006 also provides the rules on good manufacturing practices[32,33]. Furthermore, the European Food Safety Authority (EFSA) continuously conducts risk assessments and provides scientific advice on the safety of food contact materials, including those containing fluorinated substances. Another European initiative is the Registration, Evaluation, Authorisation, and Restriction of Chemicals (REACH) proposal, submitted in 2023 by the competent authorities of Germany, the Netherlands, Sweden, Norway, and Denmark. This proposal calls for a near-total ban on all PFAS substances, including fluoropolymers, across all applications by 2025. In response, the Fluoropolymers Product Group (FPG) emphasized the significant differences between low molecular weight fluorinated molecules and fluorinated polymers, highlighting the current lack of viable alternatives[34].

In the United States, the regulation of fluorinated substances in food packaging primarily falls under the jurisdiction of the US Food and Drug Administration (FDA). Following FDA indications, some long-chain fluorinated compounds such as PFOA and PFOS were voluntary phased out by the industry between 2010 and 2015 and replaced by shorter fluorinated chain analogues. Moreover, the FDA reference document, the Code of Federal Regulations (CFR), Title 21, outlines specific requirements for food contact materials, including those containing fluorinated compounds and pushes manufacturers to demonstrate that substances used in food packaging are safe and cause no adulteration of the food products[35].

Concluding remarks

In this Perspective, the use of fluorinated compounds combined with paper and paperboard for food packaging has been described in terms of fabrication techniques, classification, problems related to their extensive use both for human and environment, and current regulations in Europe and U.S.

Considering the unique water and oil repellency that fluorinated compounds can provide to paper and paperboard, further research is needed to assess whether short-chain fluorinated compounds can keep such barrier capacities and become a safer and innocuous alternative to current banned long-chain perfluorochemicals. On the other hand, modification of cellulose fibers through chemical binding with fluorinated compounds should be considered as a strategy to reduce toxicity and migration levels as well as to improve post-use degradability of food packaging.

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Competing interests

The author declares no competing interests.

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